

Hw 11

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Question 1

Use the `Cars93` dataset in the `MASS` library. Create a linear model for predicting Price from Horsepower, Weight, Length, whether a car has AirBags, and the Origin of production. Use it to predict the price of a car with 200 horsepower, weighs 2705 pounds, is 177 inches long, has driver and passenger airbags, and was not made in the USA.

```
##
## Call:
## lm(formula = Price ~ Horsepower + Weight + Length + AirBags +
##     Origin, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.9272  -2.7282  -0.2921   2.3172  27.9224
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -8.171179   10.363886  -0.788  0.43261
## Horsepower       0.104004    0.016821   6.183 2.05e-08 ***
## Weight          0.002067    0.002017   1.025  0.30825
## Length          0.044190    0.070895   0.623  0.53472
## AirBagsDriver only -3.029658   1.642779  -1.844  0.06859 .
## AirBagsNone      -6.223181   1.895908  -3.282  0.00149 **
## Originnon-USA     4.028921   1.216494   3.312  0.00136 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.37 on 86 degrees of freedom
## Multiple R-squared:  0.7111, Adjusted R-squared:  0.691
## F-statistic: 35.28 on 6 and 86 DF,  p-value: < 2.2e-16
```

Question 2

(Grad student only) Obtain regression coefficients for the model in problem 1), but using gradient descent. They should be practically equivalent. Use the `grad_descent()` and `lin_reg()` functions I have provided.

Hint: To deal with the categorical predictors, you need to make a binary column for each level that is not the default. The value should be 1 if the observation is in that category and 0 otherwise. So you can check your work, here is what the first several rows of the data matrix should look like. The tools I used are `cbind()`, `==` comparison between a dataframe column referenced by `$` and a string, and the fact that logicals can be implicitly coerced to numeric.

```
## [1] "Driver & Passenger" "Driver only"      "None"

## [1] "USA"      "non-USA"

##          1
## 30.07241

## Converged after 299 iterations.
## Local minimum of 26.57682 found at:
##          [,1]
##          1.086353e-16
## Horsepower      5.639045e-01
## Weight          1.262820e-01
## Length          6.678156e-02
## AirBags_DriverOnly -1.572344e-01
## AirBags_None      -3.119644e-01
## Origin_nonUSA      2.095671e-01
##
## Unstandardized regression coefficients:
##          Horsepower      Weight      Length
##      -8.16941272      0.10400109      0.00206784      0.04417579
## AirBags_DriverOnly      AirBags_None      Origin_nonUSA
##      -3.02980823      -6.22338126      4.02886819

##          Estimate      Std. Error      t value      Pr(>|t|)
## (Intercept)      -8.171178581 10.363886258 -0.7884280 4.326143e-01
## Horsepower      0.104003736 0.016821130 6.1829221 2.045402e-08
## Weight          0.002067368 0.002016983 1.0249805 3.082468e-01
## Length          0.044190343 0.070894535 0.6233251 5.347206e-01
## AirBagsDriver only -3.029658121 1.642778903 -1.8442276 6.859322e-02
## AirBagsNone      -6.223180743 1.895907863 -3.2824278 1.488576e-03
## Originnon-USA      4.028921352 1.216493804 3.3119128 1.356266e-03
```